Name: ______________________________ **Period:** __________ **Date:** __________

Topic 1.3 Guided Notes: The Neuron and Neural Firing

Unit 1: Biological Bases of Behavior | AP Psychology

Directions: Read through the topic overview page below and fill in each blank as you go. Use the tables and activities to check your understanding, then head to the quiz link at the end to test yourself.

Source page: appsychreviewhub.com/overviews/topic1.3

Part 1: Cellular Anatomy and the Spark of Life

Word Bank

Axon	*Sensory Neurons*	*Firing Threshold*	*Refractory Period*
Dendrites	*Action Potential*	*Soma*	*Neurons*
Multiple Sclerosis	*Resting Potential*	*Interneurons*	*Reuptake*
Motor Neurons	*All-or-Nothing Principle*	*Axon Terminal*	*Myelin Sheath*
Glial Cells	*Depolarization*	*Myasthenia Gravis*	*frequency*

Neurons and Glial Cells

The core communicators of the nervous system are ________________, specialized cells that transmit information using electrical and chemical signals. Supporting these cells are ________________, which clean up cellular waste, deliver nutrients, and build protective insulation.

The Anatomy of a Neuron — fill in each structure's name

- __________________: branch-like extensions that receive incoming signals and carry them toward the cell body — the cell's "ears."
- __________________: contains the nucleus and processes incoming information before passing it to the axon — the cell's command center.
- __________________: the long extension that carries electrical impulses away from the cell body — the transmission cable.
- __________________: a fatty insulating layer around many axons that speeds transmission of impulses.
- __________________: the end of an axon where neurotransmitters are released into the synapse — the cell's "mouth."

Diagram Labeling — Neuron Anatomy

Sketch or trace a simple neuron shape below. Using the word bank, label the five structures in the order a signal travels through them, then draw an arrow showing the direction of signal flow.

Order	Structure Name & Clue
1	(receives signal — branch-like)
2	(cell body — contains nucleus)
3	(carries impulse away from cell body)
4	(insulates the axon, speeds signal)
5	(releases neurotransmitters into synapse)

When Structures Break Down

_________________ is a disease in which the immune system damages the myelin sheath, disrupting communication and causing progressive loss of motor control. _________________ is an autoimmune disorder in which antibodies block acetylcholine receptors at the neuromuscular junction, causing muscle weakness and fatigue.

The Neural Spark: How a Neuron Fires

A neuron at rest sits at its _______________, a stable negative charge. An incoming message must push the charge to the _______________. If it does, _______________ occurs as sodium ions flood into the cell, triggering an _______________ — a brief electrical impulse traveling down the axon.

This follows the ___________________: a neuron either fires at full strength or not at all — a stronger stimulus increases firing ___________, not strength. Afterward, the neuron enters a _______________, resetting its electrical balance, and leftover neurotransmitters are reabsorbed in a process called _______________.

The Reflex Arc

Label the three neurons involved in a reflex arc, in the order information travels:

Order	Function → Neuron Type
1	Detect the stimulus and carry information toward the brain/spinal cord (afferent)
2	Located in the spinal cord; connect sensory to motor neurons
3	Carry the outgoing command to muscles/glands (efferent)

Part 2: The Whispering Chemicals (Neurotransmitters & Hormones)

Word Bank

slower	*Neurotransmitters*	*longer*
Inhibitory	*Excitatory*	*Hormones*

Neurotransmitters — Crossing the Synapse

_________________ are chemical messengers that cross the synaptic gap to carry a signal to the receiving cell — like keys that fit specific receptor "locks." _________________ neurotransmitters increase the likelihood the receiving neuron will fire, while _________________ neurotransmitters decrease that likelihood.

Match each neurotransmitter/hormone to its primary function

Chemical Messenger	Function (write the letter from the list below)
Acetylcholine	
Dopamine	
Serotonin	
Norepinephrine	
GABA	
Glutamate	
Endorphins	
Substance P	
Melatonin	
Oxytocin	
Adrenaline	
Ghrelin	
Leptin	

A) Muscle action & learning/memory

B) Movement, motivation, reward, pleasure

C) Mood, sleep, appetite

D) Alertness, arousal, stress response

E) Main inhibitory NT — calms activity

F) Main excitatory NT — learning/memory

G) Pain control & pleasure (opiate-like)

H) Transmits pain messages

I) Regulates sleep/circadian rhythm

J) Social bonding, trust, childbirth

K) Fight-or-flight — raises heart rate

L) Stimulates hunger

M) Signals fullness / enough stored energy

Hormones vs. Neurotransmitters

_________________ are chemical messengers made by the endocrine glands that travel through the bloodstream to affect other tissues. Because they travel by blood rather than jumping a synapse, they act more ___________ than neurotransmitters, but their effects linger ___________.

Write your own mnemonic

Pick two chemical messengers from this section and create your own memory trick for each (don't reuse the ones from class):

Chemical Messenger	My Mnemonic
1.	
2.	

📺 Video Notes: The Chemical Mind (Crash Course Psychology #3)

This video recaps neuron structure and firing along with today's material — only answer the questions below, which focus on the parts that add something new (neurotransmitters, reuptake, and hormones/the endocrine system).

- According to the video, what happens to "extra" neurotransmitters left in the synapse after they've done their job?
- The video mentions the endocrine system works much slower than neurons but its effects last longer. Which gland does the video call the "master gland," and why does the study guide say only this one gland matters for the AP exam?
- Pick one neurotransmitter mentioned in the video and describe, in your own words, what it does.

Part 3: Altering the Symphony (Psychoactive Drugs)

Word Bank

Depressants	*Addiction*	*Stimulants*	*Reuptake Inhibitors*
Psychoactive Drugs	*Agonists*	*Tolerance*	*Withdrawal*
Hallucinogens	*Opioids*	*Antagonists*	

Psychoactive Drugs

_________________ are chemicals that affect the central nervous system and alter mood, perception, consciousness, or behavior by influencing neurotransmitter activity.

How Drugs Interact with the Synapse — fill in the mechanism

- _________________: bind to a receptor and activate it, mimicking or amplifying a neurotransmitter's normal effect. Example: heroin acts like endorphins.
- _________________: bind to a receptor but block it, preventing the natural neurotransmitter from working. Example: Naloxone blocks opioid receptors.
- ___________________: block the transporters that reabsorb neurotransmitters, leaving more in the synapse. Example: SSRIs keep serotonin available longer.

Sort each example drug by classification and mechanism

Drug/Medication	Drug Class (Stimulant, Depressant, Hallucinogen, Opioid)	Synaptic Mechanism (Agonist, Antagonist, Reuptake Inhibitor, N-A)
Heroin		
Naloxone (Narcan)		
Cocaine		
SSRIs (e.g., Prozac)		
Caffeine		
Alcohol		
Marijuana		

The Four Core Drug Classifications

_______________ increase CNS activity (alertness, energy). _______________ slow CNS activity (reduce arousal, anxiety). _______________ alter perception and can cause hallucinations. _______________ reduce pain and produce pleasure by acting on opioid receptors.

Long-Term Neuroadaptation

- _______________: needing larger doses over time to get the same effect.
- _______________: compulsive craving of a drug or behavior despite adverse consequences.
- _______________: the discomfort/distress that follows discontinuing an addictive drug or behavior.

Quick Check: Avoid the Trap

Circle True or False, then correct any false statement on the line below it.

1. A loud scream causes a stronger action potential than a whisper. True / False

   Correction: ______________________________

2. An agonist blocks a neurotransmitter's receptor site. True / False

   Correction: ______________________________

3. Sensory neurons are efferent and motor neurons are afferent. True / False

   Correction: ______________________________

Quick Check: Review Questions

Circle the best answer for each question below.

1. When a neuron receives a signal, it uses this branch-like structure to receive incoming messages from other neurons. This structure is called the:

- **A.** Soma
- **B.** Myelin Sheath
- **C.** Dendrite
- **D.** Axon

2. A student believes that shouting loudly at a friend should cause her auditory neurons to fire a much "bigger" and more powerful electrical signal than a soft whisper would. Why is this belief incorrect?

- **A.** Louder sounds always fail to reach the neuron's firing threshold
- **B.** Only motor neurons, not sensory neurons, are capable of generating an action potential
- **C.** Neurons follow the All-or-Nothing Principle, firing at full strength every time or not at all
- **D.** Neurons only fire when a person is consciously paying attention to a stimulus

3. A patient with Parkinson's disease has abnormally low levels of a neurotransmitter that plays a major role in movement, motivation, and reward. Which neurotransmitter is deficient in this patient?

- **A.** Dopamine
- **B.** GABA
- **C.** Norepinephrine
- **D.** Endorphins

4. Naloxone (Narcan) binds to opioid receptor sites without activating them, physically blocking opioid molecules from docking there and immediately reversing an overdose. Naloxone is best classified as an:

- **A.** Stimulant
- **B.** Hallucinogen
- **C.** Agonist
- **D.** Antagonist

5. A patient taking an antidepressant experiences elevated mood because the drug blocks the transporters on the sending neuron responsible for reabsorbing serotonin, leaving more of it available in the synapse. This mechanism is known as:

- **A.** Depolarization
- **B.** The All-or-Nothing Principle
- **C.** Antagonism
- **D.** Reuptake Inhibition

6. A college student regularly drinks alcohol, which slows down her central nervous system activity, impairs her coordination, and reduces her reaction time. Alcohol is best classified as a:

- **A.** Stimulant
- **B.** Hallucinogen
- **C.** Depressant
- **D.** Opioid

✅ Check Your Progress

Done with your notes? Head over to the [Quiz](https://appsychreviewhub.com/quiz) to test what you've learned:
https://appsychreviewhub.com/quiz

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